Marmara University – Faculty of Engineering – Department of Computer Engineering

CSE4063 Fundamentals of Data Mining
Project #2

Due: 29.05.2023.Mon 23:59

1) Dataset

a) Click the link "View All Data Sets" on URL http://archive.ics.uci.edu/ml. to get the data sets assigned:

No	Group Members	Dataset [Rows x Columns]	Presentation Date
1		Absenteeism at work [740x21]	30.05.2023 **Tuesday**
2		Anuran Calls (MFCCs) [7,195x22]	
3		BLE RSSI Dataset for Indoor localization and Navigation [6,611x15]	
4		Codon usage [13,028x69]	
5		Facebook Live Sellers in Thailand [7,051x12]	
6		Estimation of obesity levels based on eating habits and physical condition [2,111x17]	30.05.2023 **Tuesday**
7		Travel Reviews [980x11]	30.05.2023 **Tuesday**
8		Mice Protein Expression [1,080x82]	
9		Gesture Phase Segmentation [9,900x50]	02.06.2023 **Friday**
10		Online Shoppers Purchasing Intention [12,330x18]	
11		Turkish Headlines Dataset [4,200x7]	06.06.2023 **Tuesday**

12		Geo-Magnetic field and WLAN dataset for indoor localisation from wristband and smartphone [153,540x25]	06.06.2023 **Tuesday** ✍
13		Motion Capture Hand Postures [78,095x38]	🕮
14		Turkiye Student Evaluation [5,820x33]	🕮
15		TV News Channel Commercial Detection [129,685x12]	09.06.2023 **Friday** ✍

b) The first students indicated by * sign are the group representatives.

c) The symbol 🕮 means:

i) You will prepare (all steps including presentation), submit, and *present* your work related to project #1.

ii) You will prepare (all steps including presentation) and submit your work related to project #2, but not present your work in class.

d) The symbol ✍ means:

i) You will prepare (all steps including presentation) and submit your work related to project #1, but not present your work in class.

ii) You will prepare (all steps including presentation), submit, and *present* your work related to project #2.

e) Learn / Get information about your data.

2) Python Platform & Environment

a) Get a platform/environment for python to work on, if you do not have any. Install it on your computer.

b) You may use any libraries you want; however, you should have complete understanding to use and explain it in demo sessions.

3) Model Construction: Frequent Pattern Mining & Clustering Analysis

a) Do the data preprocessing steps, if required.

b) Use your dataset to construct 3 frequent pattern mining models as follows:

i) Apriori.

ii) FP-Growth.

iii) ECLAT.

c) Use your dataset to construct 3 clustering analysis methods as follows:

i) K-Means.

ii) AGNES.

iii) DBSCAN.

4) Implementation & Model Evaluation

a) Implement six algorithms above on your dataset using python.

b) Compare the performance and the results of three frequent pattern mining algorithms on your dataset. Discuss the results.

c) Compare the performance and the results of three clustering analysis algorithms on your dataset. Discuss the results.

d) Compare the performances of your classifiers with performances of the relevant papers given on the site.

5) Presentation

a) You are going to present your work done online in 35 minutes at the time slot reserved for your group. Group members should equally participate the presentation. See the table above.

b) Prepare a presentation file discussing the details of your work done and results of the classifiers.

c) Your presentation should contain the following parts at least:

i) Problem definition

ii) Dataset

(1) Information about the dataset.

(2) Number of instances, columns, etc.

iii) Data preprocessing, cleaning

(1) Missing values, and how you conduct on these.

(2) Transformations and normalizations.

iv) KNIME implementation for each of the 5 algorithms

(1) IDE/environment used.

(2) Implementation details.

(3) Libraries used.

v) Model evaluation & performance results

(1) Performance measures.

(2) Comparison of all 5 algorithms.

vi) Conclusion

6) Demo with Presentation

a) You are going to demonstrate your work done online in 10 minutes after your presentation. See the table above.

b) You are going to have 35 minutes in total for your group's session (25 minutes for presentation, and 10 minutes for demonstration).

c) Please keep in mind that all the presentation and demo sessions will be recorded.

d) All the students should attend all sessions.

7) Related Questions & Answers

a) Prepare 5 questions and answers related to your topic. These questions may be asked in the exam.

b) Question types can be multiple choice (single or multiple selection), fill in the blanks, matching, essay, etc.

c) Prepare a presentation file with 11 slides consisting of these 5 questions and answers. First slide will be used for your topic and group members' info. Use 1 slide per each question, and 1 slide per each answer.

8) Evaluation

a) Your grade related to project #2 will cover 15% of your total grade.

b) Evaluation will be done out of 100 points:

i) [4 pts] Data set understanding.

ii) [4 pts] Data preprocessing.

iii) [20 pts] Implementation of frequent pattern algorithms.

iv) [20 pts] Implementation of clustering analysis algorithms.

v) [14 pts] Results, comparison, discussion & conclusion.

vi) [20 pts] Presentation quality.

vii) [8 pts] Demo quality.

viii) [10 pts] Questions & answers quality.

9) Submission

a) You are going to submit the followings:

i) Python codes/files/workflows implemented.

ii) Presentation file.

iii) Questions & answers presentation file.

b) Write the following sentence in a text file: "We hereby swear that the work done on this project is totally our own; and on our honor, we have neither given nor received any unauthorized and/or inappropriate assistance for this project. We understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues." Rename it as "we_swear.txt" and include this file in the zip submission file.

c) Only one of the group members (i.e., group representative, in short "GrRep") is going to submit the project using GrRep's info all the time. However, all group members should have a complete and comprehensive understanding of all the work done for all tasks and steps of the project.

d) Zip all your documents into a single file using filename GrRepStudentNumber_P2.zip (e.g., 150120123_P2.zip) and submit it to the site http://ues.marmara.edu.tr before deadline.

e) In case of any form of copying and cheating on solutions, all parts will get ZERO points. You should submit your own work. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties. All types of plagiarism will result in zero points from the homework.

f) If case of using your handwriting, your handwriting should be readable, clear, and neat. If possible, do not use any handwriting.

g) Do not send project submissions through e-mail. E-mail attachments will not be accepted as valid submissions.

h) You are responsible for making sure you are turning in the right file, and that it is not corrupted in anyway. We will not allow resubmissions if you turn in the wrong file, even if you can prove that you have not modified the file after the deadline.

i) Grade evaluation may be done on selected parts of the project, so try to complete all parts of your project successfully.

j) No late submissions will be accepted.